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Prof. Kenneth M. Merz, Jr., Editor-in-Chief
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Dear Prof. Merz,

We submit our manuscript **"Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis"** for consideration as an Article in *Journal of Chemical Information and Modeling*.

This work addresses a critical methodological challenge in computational antimalarial drug discovery: evaluating target breadth and mutation resilience as distinct, complementary properties when prioritizing candidates from African natural-product-inspired chemical space, while maintaining clear boundaries between computational evidence and biological activity. We analyze a 17-member polypharmacology-oriented cohort across four mechanistically diverse antimalarial targets (PfDHFR, PfCRT, PfClpP, and PfATP4) using target-specific structural anchors and a per-target resistance-resilience scoring (RRS) framework. The workflow integrates three evidence layers: chemical-space expansion (65 856 hybrid library molecules $\rightarrow$ 19 913 synthesizable candidates), target-anchored AutoDock Vina docking generating a complete 17×4 scoring matrix (68/68 geometric quality pass rate), and per-target RRS calculated separately for PfDHFR (N51I, C59R, S108N, I164L) and PfCRT (K76T, K76A) mutant panels. The cohort yielded six A*, five B, five C, and one D RRS profiles (range 68.2–111.7), with weak cross-metric associations after multiplicity correction (PNS–RRS $\rho = -0.559$, ACSI–RRS $\rho = -0.132$).

Key enhancements (comprehensive validation, physicochemical characterization, and methodological documentation):

1. **Validation documentation:** three complementary approaches: (a) external benchmark (DEKOIS 2.0, PfDHFR ROC-AUC 0.45, reported honestly as near-chance), (b) enrichment validation (MMV Malaria Box, ROC-AUC 0.924–1.000), and (c) redocking validation (four of five reference ligands reproduced with RMSD < 2.0 Å; the methotrexate exception reflects NADPH-adjacent grid placement). Metrics in Supporting Information Section S12.
2. **Physicochemical characterization:** all 17 candidates exhibit drug-like properties (100 % Lipinski/Veber compliance, mean QED 0.703, low predicted ADMET risk). Details in Supporting Information Tables S2–S4.
3. **Multi-parameter optimization framework:** a five-component MPO (Vina affinity 35%, DiffDock confidence 25%, QED 20%, ADMET 15%, Lipinski penalty 5%) with a polypharmacology bonus and an MPO $\geq$ 0.40 selection threshold (Section S10).
4. **Methodological rigor:** complete grid box specifications for all four targets, 99.3 % cost reduction via centroid-based library reduction, and strict evidence boundaries.

Evidence boundaries: the study does not claim experimental affinity, confirmed simultaneous target engagement, biological resistance circumvention, or measured IC_{50}/EC_{50} values. The workflow produces (a) four-target docking profiles from which polypharmacology hypotheses can be generated, (b) per-target RRS values summarizing mutation-panel docking ratios, and (c) exploratory cross-metric associations without conflation.

Reproducibility and impact: complete computational provenance (source code, candidate manifests, receptor/ligand files, grid configurations, raw poses, RRS protocols, machine-readable tables) is archived at `https://github.com/NanaEngo/Malaria_codesV2`; the Supporting Information (17 pages, 12 sections, 9 tables, 2 figures) contains the validation datasets and methodological specifications.

The centroid-based reduction strategy lowers the computational barrier for endemic-region groups (94 % of malaria cases) by orders of magnitude while retaining structural diversity, providing a reproducible template for hypothesis generation in resistance-challenged therapeutic areas. This manuscript is original, not under consideration elsewhere, and all authors have reviewed and approved the submission. We declare no competing financial interests.

Sincerely,
Myke Vital Sao Temgoua (on behalf of all co-authors)